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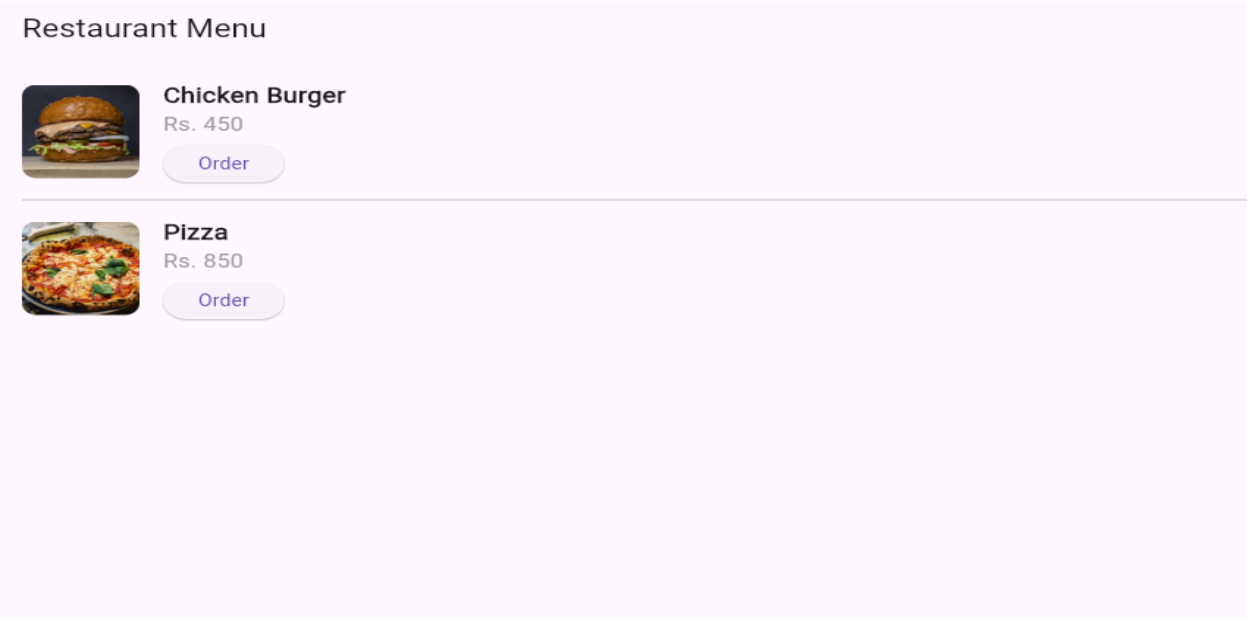
Teacher:Sir Usman

Subject: Mobile Application

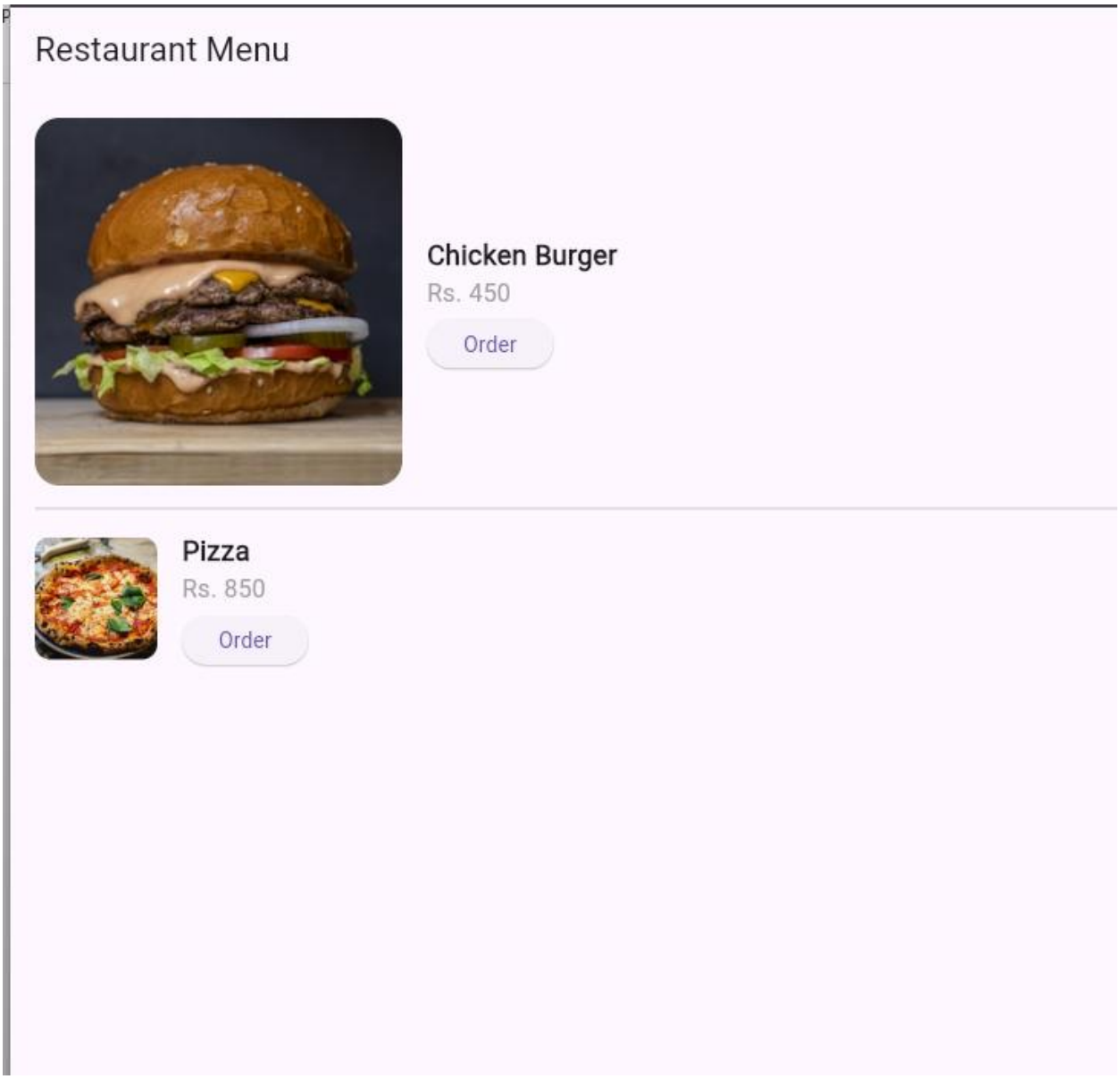
Animation Task

Task 1

Before



After



Container

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
```

```
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      home: MenuPage(),
    );
  }
}

class MenuPage extends StatelessWidget {
  const MenuPage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Restaurant Menu'),
      ),
      body: SafeArea(
        child: SingleChildScrollView(
          padding: const EdgeInsets.all(16.0),
          child: Column(
            children: const [
              FoodItem(
                name: 'Chicken Burger',
                price: 'Rs. 450',
                imageUrl: 'https://images.unsplash.com/photo-1568901346375-23c9450c58cd?w=300',
              ),
              Divider(height: 30),
              FoodItem(
                name: 'Pizza',
                price: 'Rs. 850',
                imageUrl: 'https://images.unsplash.com/photo-1574071318508-1cdbab80d002?w=300',
              ),
            ],
          ),
        ),
      ),
    );
```

```
  }
}

class FoodItem extends StatefulWidget {
  final String name;
  final String price;
  final String imageUrl;

  const FoodItem({
    super.key,
    required this.name,
    required this.price,
    required this.imageUrl,
  });

  @override
  State<FoodItem> createState() => _FoodItemState();
}

class _FoodItemState extends State<FoodItem> {
  bool _crossFadeStateShowFirst = true;

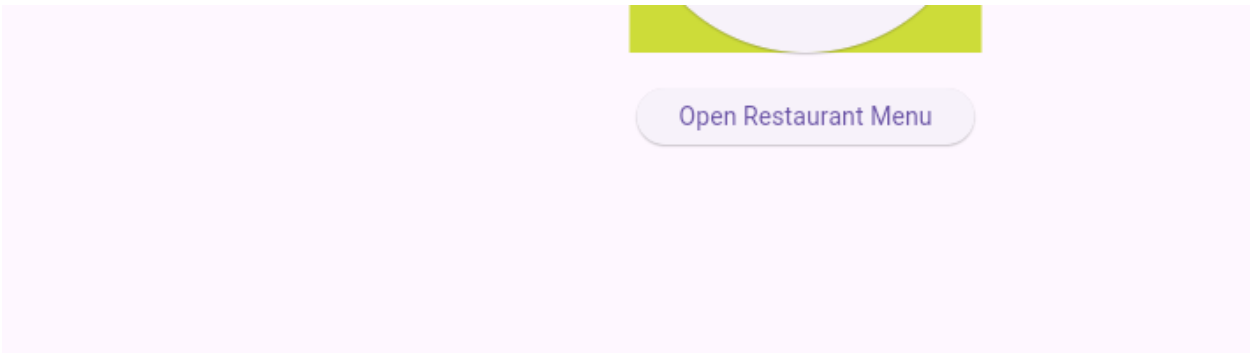
  void _crossFade() {
    setState(() {
      _crossFadeStateShowFirst = !_crossFadeStateShowFirst;
    });
  }

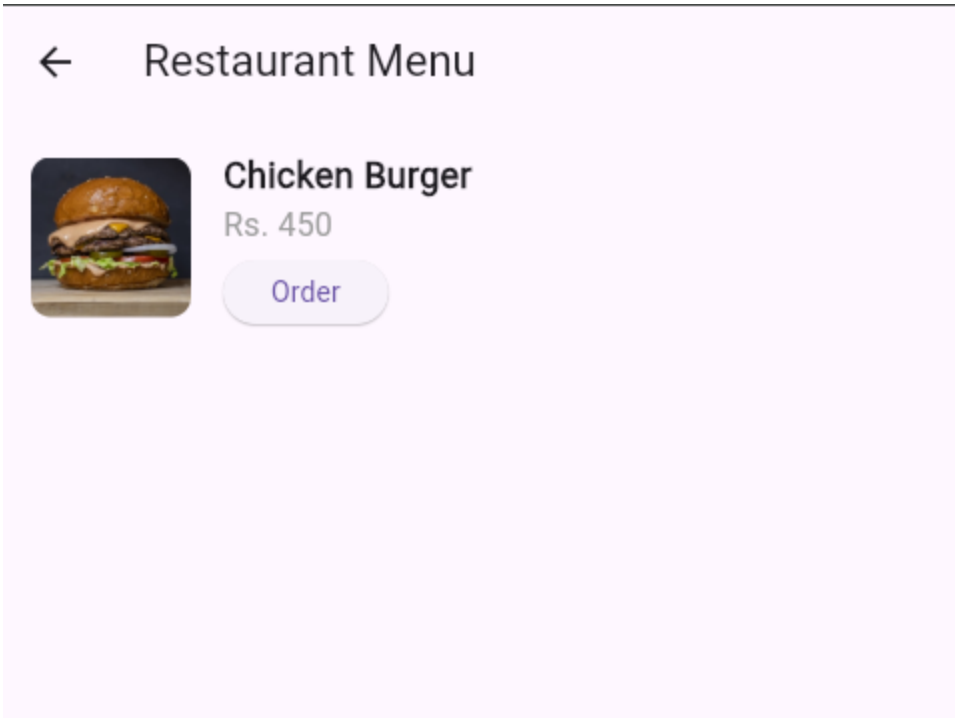
  @override
  Widget build(BuildContext context) {
    return Row(
      crossAxisAlignment: CrossAxisAlignment.center,
      children: [
        // Using AnimatedCrossFade with balanced dimensions
        GestureDetector(
          onTap: _crossFade,
          child: AnimatedCrossFade(
            duration: const Duration(milliseconds: 500),
            sizeCurve: Curves.bounceOut,
            crossFadeState: _crossFadeStateShowFirst
                ? CrossFadeState.showFirst
                : CrossFadeState.showSecond,
            firstChild: ClipRRect(
              borderRadius: BorderRadius.circular(10),
```

```
      child: Image.network(
        widget.imageUrl,
        width: 80,
        height: 80,
        fit: BoxFit.cover,
      ),
    ),
    secondChild: ClipRRect(
      borderRadius: BorderRadius.circular(16),
      child: Image.network(
        widget.imageUrl,
        width: 240,
        height: 240,
        fit: BoxFit.cover,
      ),
    ),
  ),
),
const SizedBox(width: 16),
Expanded(
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Text(
        widget.name,
        style: const TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
      ),
      Text(
        widget.price,
        style: const TextStyle(fontSize: 16, color: Colors.grey),
      ),
      const SizedBox(height: 6),
      ElevatedButton(
        onPressed: () {
          ScaffoldMessenger.of(context).showSnackBar(
            const SnackBar(
              content: Text('Order successfully!'),
              duration: Duration(seconds: 5),
            ),
          );
        },
        child: const Text('Order'),
      ),
    ],
```

```
        ),
      ),
    ],
  );
 }
}
```

Task 2






```
import 'package:flutter/material.dart';
import 'AnimatedContainerWidget.dart';
import 'AnimatedCrossFadeWidget.dart';
import 'MenuPage.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});
```

```
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'My First App',
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('My First App'),
        centerTitle: true,
      ),
      body: SafeArea(
```

```
        child: SingleChildScrollView(
          child: Column(
            crossAxisAlignment:
CrossAxisAlignment.center,
            children: <Widget>[
              const Text('Hello World !!!'),
              const SizedBox(height: 10),

              Container(
                width: 100,
                height: 100,
                color: Colors.amber,
              ),

              const SizedBox(height: 20),

              const AnimatedContainerWidget(),

              const SizedBox(height: 20),

              const AnimatedCrossFadeWidget(),
```

```
            const SizedBox(height: 20),

            ElevatedButton(
              onPressed: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(builder: (_) => const
MenuPage()),
                );
              },
              child: const Text('Open Restaurant
Menu'),
            ),

            const SizedBox(height: 20),
          ],
        ),
      ),
    ),
  );
 }
}
```